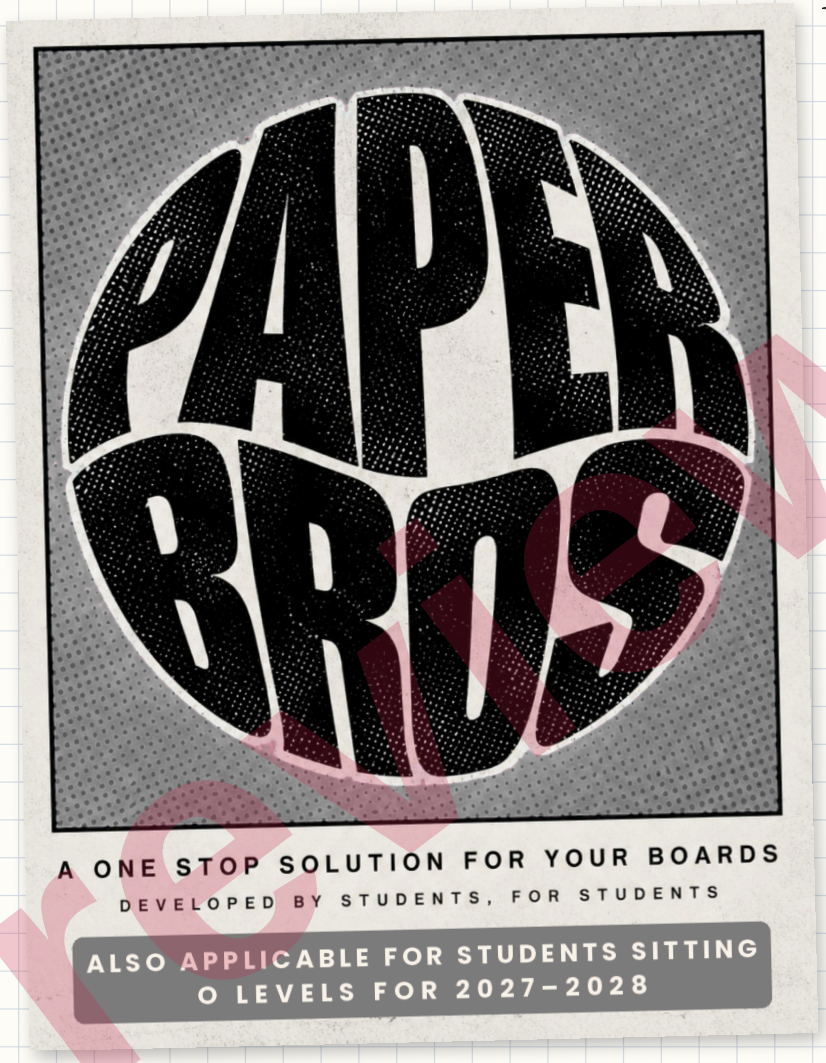
PREVIEW

Cambridge O Level

Physics 5054

the complete notes – for exams in 2029



notes for every syllabus point · worked examples
accurate diagrams & graphs · Paper 1 & Paper 2 practice + answers

this notebook belongs to:

Name .....

School / class ..... Exam: June / Nov

Preview



what's inside

Contents



tick the box when a topic is done & dusted ✓

#	Topic	What's inside (syllabus sections)	Notes pg	Practice pg	✓
–	The exam at a glance	papers, marks, AOs, command words, units & prefixes, how to use these notes	3	–	☐
–	Every equation you must know	all the "recall and use" equations on one page (nothing is given in the exam!)	4	–	☐
1A	Measurement, motion & density	1.1 measurement, scalars & vectors · 1.2 motion & graphs · 1.3 mass & weight · 1.4 density	5	9	☐
1B	Forces & momentum	1.5 forces, Newton's laws, friction & stopping, springs, circular motion, moments, centre of gravity · 1.6 momentum	12	17	☐
1C	Energy, work, power & pressure	1.7 energy stores, work, energy resources, efficiency, power · 1.8 pressure	20	24	☐
2	Thermal physics	2.1 particle model, gas pressure · 2.2 expansion, specific heat capacity, melting/boiling/evaporation · 2.3 conduction, convection, radiation	27	31	☐
3A	Waves & light	3.1 general wave properties, ripple tank · 3.2 reflection, refraction, lenses, dispersion	34	39	☐
3B	EM spectrum & sound	3.3 electromagnetic spectrum, uses & dangers · 3.4 sound, echoes, ultrasound	42	44	☐
4A	Magnetism & electrical quantities	4.1 magnets & fields · 4.2 charge, current, e.m.f., p.d., resistance	46	50	☐
4B	Circuits & practical electricity	4.3 circuit symbols, series & parallel, sensors, potential dividers · 4.4 power, kW h, electrical safety	53	57	☐
4C	Electromagnetic effects	4.5 induction, generator, relay, motor effect, d.c. motor, transformer · 4.6 oscilloscope	60	64	☐
5	Nuclear physics	5.1 atom & nucleus · 5.2 detection, α β γ, decay equations, fission & fusion, half-life, safety	67	71	☐
6	Space physics	6.1 Earth, Moon & Solar System · 6.2 the Sun, stars & their life cycle, the Universe, redshift	74	77	☐
✶	Answers	Paper 1 keys + Paper 2 mark-scheme style answers for every topic	80	–	☐

Same content as 2027 & 2028 ✓

Cambridge says the 2029 syllabus has **no significant changes which affect teaching** – so these notes work for the June/Nov 2027 and 2028 series too. Textbooks endorsed from 2023 are still fine.

How I'd use this book

1. read the notes → 2. cover them & redo the worked examples → 3. do the practice without peeking → 4. mark with the answers at the back & fix your mistakes in red.

Topics 1 and 4 are huge, so they're split into parts (A, B, C) – each part has its own practice questions.

Preview

PREVIEW

before we start...

The Exam at a Glance

You sit **three** papers. Everyone takes Paper 1 **and** Paper 2, then **either** Paper 3 **or** Paper 4. Grades go from **A*** down to **E**.

Paper	Time	Marks	Weight	What it looks like	Tests
1 Multiple Choice	1 h	40	30%	40 compulsory four-option (A–D) questions	AO1 + AO2
2 Theory	1 h 45 min	80	50%	compulsory short-answer & structured questions	AO1 + AO2
3 Practical Test	1 h 30 min	40	20%	you do real experiments in a lab	AO3
		– OR –			
4 Alternative to Practical	1 h	40	20%	written questions about experiments (no lab)	AO3

AO Assessment objectives

Objective	Weight
AO1 Knowledge with understanding – laws, definitions, equations, units, apparatus	50%
AO2 Handling information & problem-solving – data, graphs, calculations, new situations	30%
AO3 Experimental skills & investigations – planning, measuring, evaluating	20%

Inside Papers 1 & 2 the split is roughly **63% AO1 : 37% AO2**: over a third of the marks are "use it somewhere new" – so practise!

no formula sheet! ⚠
Unlike some exams, Physics 5054 gives you **no list of equations**. Every "recall and use" equation must come from your head – they're all on the next page.

✱ Prefixes you must use [syllabus p.42]

prefix	G	M	k	d	c	m	μ	n
name	giga	mega	kilo	deci	centi	milli	micro	nano
×	10^9	10^6	10^3	10^{-1}	10^{-2}	10^{-3}	10^{-6}	10^{-9}

conversions that trip people up 1 m^3 = 1 000 000 cm^3 · 1 cm^2 = 0.0001 m^2 · 1 g/cm^3 = 1000 kg/m^3 · 72 km/h = 20 m/s (÷ 3.6)

key How to read these notes

- **maroon bold** = a key word that earns the mark
- yellow highlight = learn word-for-word
- boxed equation = recall it & use it
- yellow box = **definition**; blue box = **worked example**
- sticky notes = exam tips & traps ⚠
- little grey numbers like [1.5.3] = the syllabus point

✱ Practical words (Papers 3 & 4)

accurate	close to the **true value**
precise	repeated values are **close to each other**
independent / dependent variable	the one **you change** / the one **you measure**
anomaly	a result outside the general pattern – repeat it or leave it out of the mean
reading a scale	to the nearest **half** of the smallest division

cmd Command words

The first word of the question tells you what the examiner wants:

Word	What you do
State / Give	short clear answer, no explanation needed
Define	the precise meaning – learn definitions word-for-word
Describe	the main points / features (what you'd see)
Explain	give reasons *why* or *how* – use "because..." and link the steps
Calculate	work out a number – equation, substitution, answer + unit
Determine	get an answer using data / a graph given
Deduce	conclude from the information given
Suggest	apply ideas to a new situation – several answers can be right
Predict	say what may happen using the information
Compare	similarities and/or differences – mention both things
Comment	give an informed opinion
Identify	name / select / recognise
Justify	support a case with evidence
Sketch	simple freehand drawing (or graph) showing the key features & correct shape

✱ The 4-line method for every calculation

WE ✎ how to set it out

A car travels 150 m in 6.0 s. Calculate its average speed.

① equation: $v = s \div t$
② substitute: $v = 150 \div 6.0$
③ answer + unit: $v = 25$ m/s
④ sense check: 25 m/s ≈ 90 km/h – a fast car ✓

Top 3 mark-losers
① missing or wrong **units** (N vs kg!) ② forgetting to convert **cm** ⟶ **m**, **g** ⟶ **kg**, **min** ⟶ **s** ③ rounding too early – only round the final answer (2–3 s.f.).

Preview

PREVIEW

learn these by heart

Every Equation You Must Know

cover the middle column & test yourself every week

in words	symbols	units
1 · Motion, forces & energy		
speed = distance ÷ time	$v = \frac{s}{t}$	m/s
average speed	$\frac{\text{total distance}}{\text{total time}}$	m/s
acceleration	$a = \frac{\Delta v}{\Delta t}$	m/s^2
gravitational field strength	$g = \frac{W}{m}$	N/kg
density	$\rho = \frac{m}{V}$	kg/m^3, g/cm^3
resultant force	$F = ma$	N
spring constant	$k = \frac{F}{x}$	N/m, N/cm
moment	force × ⊥ distance from pivot	N m
momentum	$p = mv$	kg m/s
impulse	$F\Delta t = \Delta(mv)$	N s
resultant force (momentum form)	$F = \frac{\Delta p}{\Delta t}$	N
kinetic energy	$E_k = \frac{1}{2}mv^2$	J
change in gravitational p.e.	$\Delta E_p = mg\Delta h$	J
work done	$W = Fd$	J
efficiency	$\frac{\text{useful output}}{\text{total input}} \times 100\%$	%
power	$P = \frac{W}{t} = \frac{\Delta E}{t}$	W
pressure	$p = \frac{F}{A}$	Pa = N/m^2
pressure change in a liquid	$\Delta p = \rho g \Delta h$	Pa
2 · Thermal physics		
gas at constant temperature	$p_1V_1 = p_2V_2$	any matching units
kelvin ↔ Celsius	$T = \theta + 273$	K
specific heat capacity	$c = \frac{\Delta E}{m\Delta\theta}$	J/(kg °C)

in words	symbols	units
3 · Waves		
wave speed	$v = f\lambda$	m/s
refractive index	$n = \frac{\sin i}{\sin r}$	no unit
critical angle	$n = \frac{1}{\sin c}$	no unit
linear magnification	$\frac{\text{image length}}{\text{object length}}$	no unit
4 · Electricity & magnetism		
current	$I = \frac{Q}{t}$	A = C/s
e.m.f. (work by source)	$E = \frac{W}{Q}$	V = J/C
p.d. (work on component)	$V = \frac{W}{Q}$	V = J/C
resistance	$R = \frac{V}{I}$	Ω
wire resistance	$R \propto$ length, $R \propto \frac{1}{\text{area}}$	Ω
resistors in series	$R = R_1 + R_2 + \ldots$	Ω
two resistors in parallel	$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2}$	Ω
potential divider	$\frac{R_1}{R_2} = \frac{V_1}{V_2}$	–
electrical power	$P = IV$	W
electrical energy	$E = IVt$	J (or kW h)
transformer	$\frac{V_p}{V_s} = \frac{N_p}{N_s}$	–
6 · Space physics		
average orbital speed	$v = \frac{2\pi r}{T}$	m/s, km/s

✱ Numbers worth knowing

g near Earth	≈ 9.8 m/s^2 = 9.8 N/kg
speed of light (EM waves) in vacuum	3.0 × 10^8 m/s
speed of sound in air	330–350 m/s
human hearing	20 Hz – 20 000 Hz
absolute zero	–273 °C = 0 K
water melts / boils	0 °C / 100 °C
1 kW h	3 600 000 J = 3.6 MJ
Sun's light to Earth	≈ 500 s

rearranging made easy ▲

For any **A = B × C**: cover the one you want – B = A ÷ C, C = A ÷ B. So $F = ma$ gives $a = F/m$; $v = f\lambda$ gives $\lambda = v/f$; $P = IV$ gives $I = P/V$. Always rearrange **before** you put numbers in.

graph shortcuts 📈

gradient of distance–time = **speed** · gradient of speed–time = **acceleration** · area under speed–time = **distance**

Preview

Topic 1 · Part A

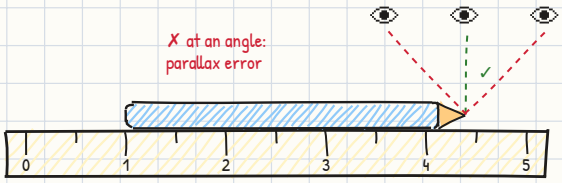
Measurement, Motion & Density

how far, how fast, how heavy — the foundations of everything

1.1 Physical quantities & measurement

✱ Measuring lengths [1.1.1]

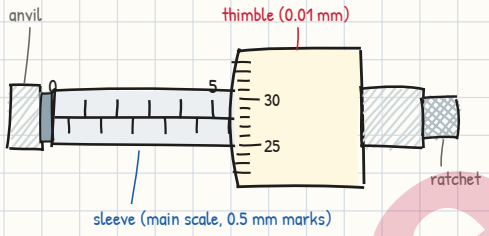
instrument	use for	precision
tape measure	long lengths (room, running track)	1 mm
metre rule / ruler	lengths from a few cm to 1 m	1 mm
micrometer	tiny lengths (wire diameter, paper thickness)	0.01 mm



object starts at the 1 cm mark ⇒ length = 4.4 − 1.0 = 3.4 cm (not 4.4!)

Eye **directly above** the mark (no **parallax error**). Start at a mark, not the worn end, and subtract.

✱ Reading a micrometer



WE ✎ read it

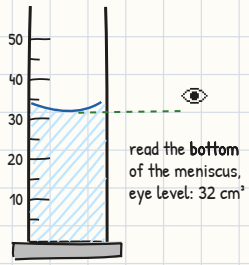
sleeve: last visible mark = **5.5 mm** (the half-mm line below the datum is showing)

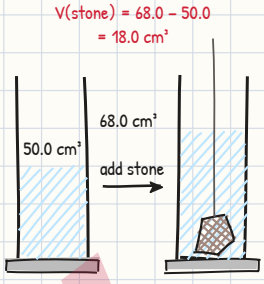
thimble: line **28** meets the datum line ⇒ 28 × 0.01 = **0.28 mm**

reading = 5.5 + 0.28 = **5.78 mm**

zero error close the micrometer with nothing in it first. If it reads +0.02 mm, **subtract** 0.02 mm from every reading. Turn the **ratchet** so you don't squash the object.

✱ Volumes [1.1.2]



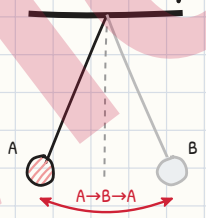


The volume of an **irregular solid** = the rise in liquid level when it's lowered in (gently, on a thread) = volume by displacement. It must sink and not dissolve.

✱ Time [1.1.3]

stop-watch / clock (reading to 0.01 s, but your **reaction time** ≈ 0.2 s is the real limit) · digital timers with **light gates** remove reaction time completely.

✱ Measure many, then divide [1.1.4]



One **oscillation** = A ⟶ B ⟶ back to A. Time **20** and divide: your reaction-time error is shared over 20 swings ⇒ much smaller **percentage** error.

WE ✎ period & thickness

20 oscillations take 32.4 s ⇒ period T = 32.4 ÷ 20 = **1.62 s**

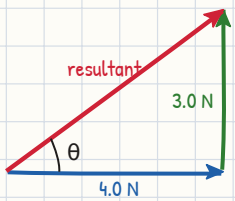
a stack of 250 sheets is 2.5 cm thick ⇒ 1 sheet = 2.5 ÷ 250 = **0.010 cm = 0.10 mm**

✱ Scalars & vectors [1.1.5–8]

scalar = has magnitude (size) only · **vector** = has magnitude and direction

scalars	vectors
distance, speed, time, mass, energy, temperature	displacement, force, weight, velocity, acceleration, momentum, electric field strength, gravitational field strength

trick if "which way?" makes sense, it's a vector. "10 kg north"? nonsense ⇒ scalar.



Two vectors at right angles: draw them **tip-to-tail**; the resultant joins start to finish. Calculate (Pythagoras + tan) or use a **scale drawing**.

WE ✎ 4.0 N east + 3.0 N north

size = $\sqrt{(4.0^2 + 3.0^2)}$ = $\sqrt{25}$ = **5.0 N**

$\tan\theta$ = 3.0 ÷ 4.0 ⇒ θ = **37° north of east**

Preview

1.2 Motion

speed = distance travelled per unit time
velocity = change in displacement per unit time (speed in a given direction) [1.2.1]

$$\text{speed} = \frac{\text{distance}}{\text{time}} \qquad v = \frac{s}{t}$$

$$\text{average speed} = \frac{\text{total distance travelled}}{\text{total time taken}}$$

WE 1 ✎ average speed [1.2.3]

A bus travels 12 km in 15 min, waits 5 min at a stop, then travels 8 km in 10 min. Find its average speed in m/s.
distance = 12 + 8 = **20 000 m**; time = 15 + 5 + 10 min = **1800 s** (the stop counts!)
average speed = 20 000 ÷ 1800 = **11 m/s** (11.1)

speed ≠ velocity

A horse on a merry-go-round moves at a **constant speed**, but its **direction keeps changing** ⇒ its velocity is changing ⇒ it is accelerating! Velocity = speed + **direction**.

acceleration = change in velocity per unit time [1.2.4]

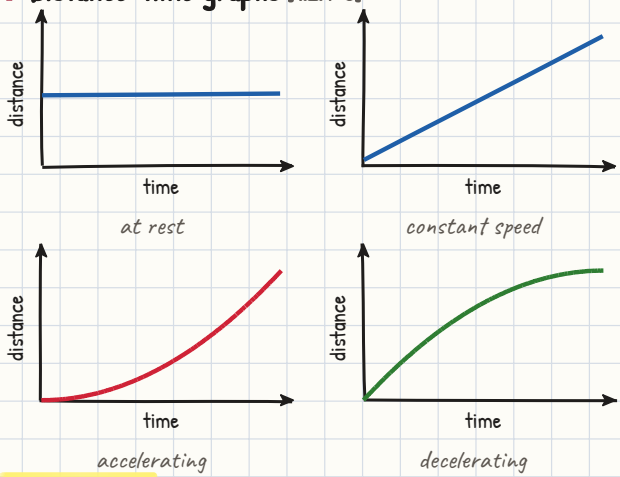
$$\text{acceleration} = \frac{\text{change in velocity}}{\text{time taken}} \qquad a = \frac{\Delta v}{\Delta t} \quad \text{m/s}^2$$

- **uniform** acceleration: velocity changes by the **same amount every second** (a falling stone, no air resistance) [1.2.5]
- **non-uniform**: the rate keeps changing (a car pulling away and changing gear, a skydiver)
- a **deceleration** is a **negative acceleration** [1.2.6]

WE 2 ✎ braking

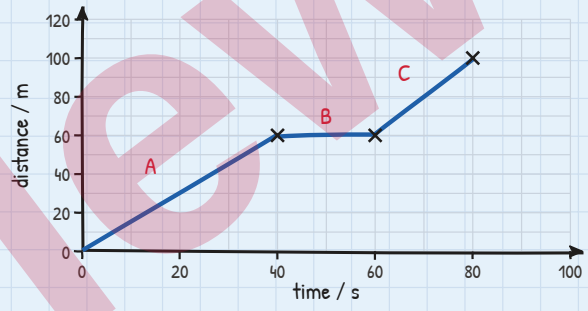
A car slows from 25 m/s to 5.0 m/s in 4.0 s.
$a = (5.0 - 25) \div 4.0 =$ **-5.0 m/s²** (a deceleration of 5.0 m/s²)

✱ Distance–time graphs [1.2.7–8]



gradient = speed. Straight line = constant speed; steeper = faster; curving **up** = speeding up; levelling off = slowing down; flat = stopped. [1.2.11]

WE 3 ✎ a walk to the shop



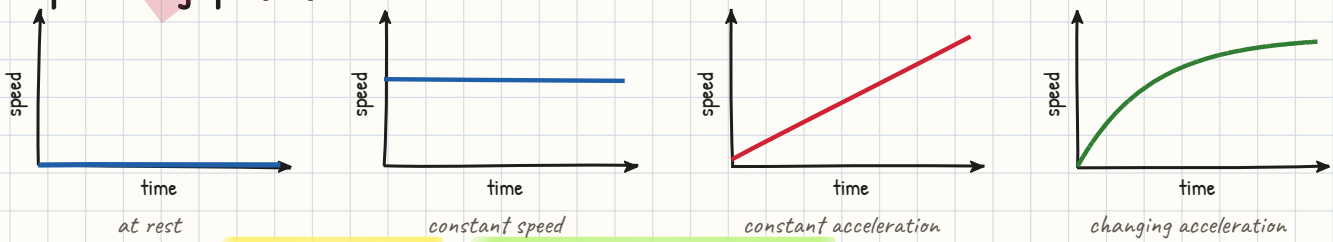
A: 60 m ÷ 40 s = **1.5 m/s** B: flat ⇒ **at rest** for 20 s
C: (100 − 60) ÷ (80 − 60) = 40 ÷ 20 = **2.0 m/s** (steeper ⇒ faster ✓)
average speed = 100 m ÷ 80 s = **1.25 m/s**

curved line? draw a **tangent** at that point and find its gradient — that's the speed at that instant.

how fast is Bolt? ⚡

100 m in 9.58 s ⇒ **average** 10.4 m/s, but his **top** speed was about 12 m/s — he was slower at the start.

✱ Speed–time graphs [1.2.9]

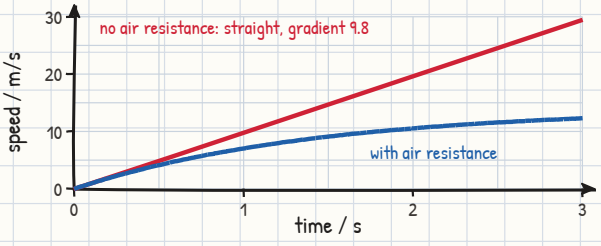


gradient = acceleration · **area under the line = distance travelled** · sloping down = deceleration

✱ Free fall [1.2.10]

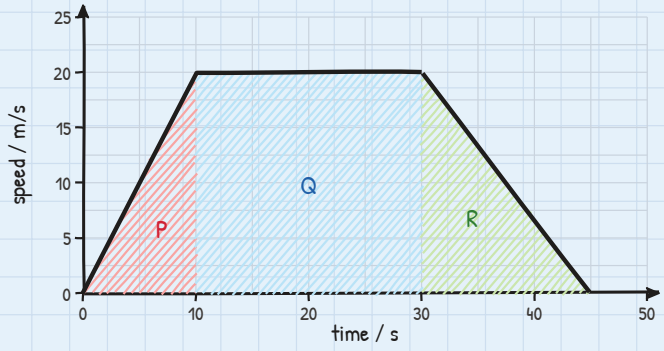
Near the Earth's surface, the **acceleration of free fall** g is approximately constant: $g \approx 9.8$ m/s²

Without air resistance **every** object falls with the same acceleration — on the Moon a hammer and a feather land together.



Preview

WE ✎ the big one — a car journey on a speed–time graph [1.2.12–13]



(a) acceleration in P: $a = (20 - 0) \div 10 =$ **2.0 m/s²**

(b) acceleration in R: $(0 - 20) \div (45 - 30) =$ **−1.3 m/s²**

(c) distance = total area:

P (triangle) = ½ × 10 × 20 = 100 m

Q (rectangle) = 20 × 20 = 400 m

R (triangle) = ½ × 15 × 20 = 150 m

total = **650 m**

(d) average speed = 650 ÷ 45 = **14 m/s** (14.4)

graph traps ⚠

① a **horizontal** line means constant *speed* on a speed–time graph but *at rest* on a distance–time graph — check the axis label first! ② areas are in **graph units**: if time is in minutes, convert to seconds. ③ "gradient" means rise ÷ run using the **axis numbers**, not centimetres on the paper.

1.3 Mass & weight

mass = a measure of the **quantity of matter** in an object (at rest relative to the observer). Unit: **kg**. [1.3.1]

inertia: mass **resists change** from its state of rest or motion. [1.3.2] A loaded trolley is harder to start *and* harder to stop than an empty one.

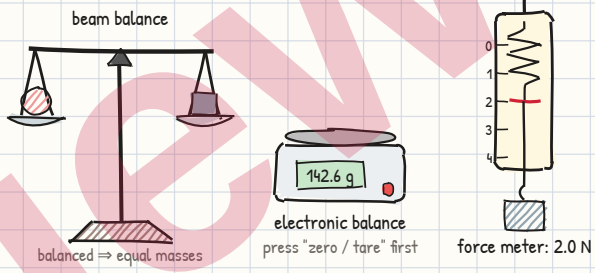
weight = the **gravitational force** on an object that has mass. Unit: **N**. It is a **vector** (acts downwards).

gravitational field strength = **force per unit mass** [1.3.6]
gravitational field = a region in which a mass experiences a force due to gravitational attraction [1.3.7]

$g = \frac{W}{m}$ so $W = mg$ — g on Earth = 9.8 N/kg

g in N/kg is **the same thing** as the acceleration of free fall in m/s² (1 N/kg = 1 m/s²).

	mass	weight
what it is	amount of matter	force of gravity on it
unit	kg	N
scalar / vector	scalar	vector (downwards)
on the Moon	same	about ⅙ as big
measured with	balance	force meter (newton meter)



[1.3.3–5] A **beam balance** compares weights (and so masses). An **electronic balance**: zero it, place the object on the pan, read the mass. A **force meter** (spring) measures weight: hang the object and read at eye level.

WE ✎ astronaut on the Moon

An astronaut has mass 70 kg. g_{Earth} = 9.8 N/kg, g_{Moon} = 1.6 N/kg.

weight on Earth = 70 × 9.8 = **690 N** (686)

weight on Moon = 70 × 1.6 = **110 N** (112)

mass on the Moon = **70 kg** — mass never changes!

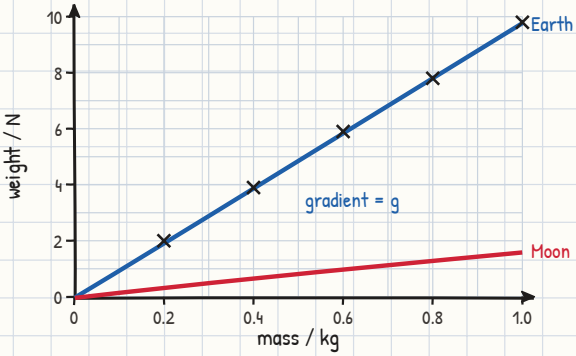
"I weigh 50 kg" ✕

In physics that's wrong — your **mass** is 50 kg; your **weight** is about 490 N. Bathroom scales secretly measure force and divide by 9.8!

✱ Experiment: weight vs mass

1. hang a 100 g mass on a force meter; read the weight at eye level
2. add masses 100 g at a time up to 1000 g, recording each weight
3. convert g ⟶ kg and plot **weight / N** against **mass / kg**

Result: a **straight line through the origin** ⇒ weight ∝ mass. The **gradient = g** = 9.8 N/kg.



WE ✎ from the graph

gradient of the Earth line = 9.8 ÷ 1.0 = **9.8 N/kg**; Moon line = 1.6 ÷ 1.0 = **1.6 N/kg**

Preview

1.4 Density

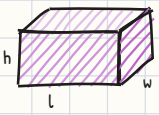
density = mass per unit volume [1.4.1]

$$\text{density} = \frac{\text{mass}}{\text{volume}} \qquad \rho = \frac{m}{V}$$

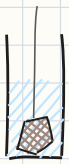
units: g/cm³ or kg/m³ · 1 g/cm³ = 1000 kg/m³ (water!)

substance	density / kg/m³	substance	density / kg/m³
air	1.2	aluminium	2700
cork	240	iron / steel	7900
ice	920	copper	8900
water	1000	mercury	13 600
sea water	1030	gold	19 300

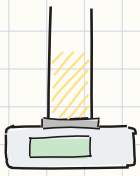
✱ Three ways to find density [1.4.2]



① regular solid
V = l × w × h (ruler)



② irregular solid
V by displacement



③ liquid
mass with − mass empty

- always find the **mass on a balance** first (in g) — then the volume (in cm³)
- for ③: weigh the empty measuring cylinder, pour in liquid, read the volume, weigh again; **mass of liquid = difference**

WE 4 ✎ an irregular stone

A stone of mass 54 g is lowered into water; the level rises from 50 cm³ to 70 cm³.

V = 70 − 50 = 20 cm³ · ρ = 54 ÷ 20 = 2.7 g/cm³

Eureka!

Legend says Archimedes noticed the bath overflow as he got in — and realised he could find the volume of the king's crown by displacement, then its density, to check if it was pure gold (19 300 kg/m³).

WE 1 ✎ regular block

A block is 5.0 cm × 4.0 cm × 2.0 cm; mass 108 g. What metal?

V = 5.0 × 4.0 × 2.0 = 40 cm³ · ρ = 108 ÷ 40 = 2.7 g/cm³

= 2700 kg/m³ ⇒ it's **aluminium** (check the table)

WE 2 ✎ a liquid

empty cylinder 62.0 g; with 50.0 cm³ of liquid 101.5 g

mass of liquid = 101.5 − 62.0 = 39.5 g

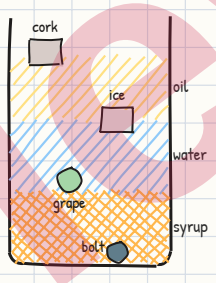
ρ = 39.5 ÷ 50.0 = 0.79 g/cm³ = 790 kg/m³

WE 3 ✎ rearranging

What is the mass of 0.020 m³ of steel (ρ = 7900 kg/m³)?

$m = \rho V$ = 7900 × 0.020 = 158 kg

✱ Float or sink?



An object **floats** in a liquid if its density is **less** than the liquid's, and **sinks** if greater. Liquids that don't mix form layers, **densest at the bottom**.

but steel ships float?!

Steel (7900 kg/m³) sinks — but a ship is mostly **air inside**. Its **average** density (steel + air) is less than sea water's 1030 kg/m³. Fill it with water ⇒ average density rises ⇒ it sinks.

✱ Part A in one picture

measure
micrometer: 0.01 mm
eye level, no parallax
time 20, divide by 20
scalar: size only
vector: size + direction

motion
v = s ÷ t
a = Δv ÷ Δt (m/s²)
deceleration = negative a
g ≈ 9.8 m/s²
velocity has direction

graphs
d-t gradient = speed
v-t gradient = accel.
v-t area = distance
flat d-t = at rest
curve: draw a tangent

mass, weight, density
W = mg (N)
mass same everywhere
ρ = m ÷ V
1 g/cm³ = 1000 kg/m³
less dense ⇒ floats

Preview



Topic 1A · measure twice, answer once

Practice Questions



PAPER 1 · MULTIPLE CHOICE

1 Which instrument is best for measuring the diameter of a thin wire?

A metre rule B micrometer
C measuring cylinder D tape measure

2 A pendulum makes 20 complete oscillations in 36.0 s. What is its period?

A 0.56 s B 1.8 s C 18 s D 720 s

3 Which list contains only vector quantities?

A distance, speed, time
B mass, weight, energy
C force, velocity, acceleration
D displacement, temperature, momentum

4 Forces of 6.0 N (east) and 8.0 N (north) act on an object. What is the size of the resultant force?

A 2.0 N B 7.0 N C 10 N D 14 N

5 A car accelerates uniformly from 10 m/s to 30 m/s in 5.0 s. What is its acceleration?

A 2.0 m/s^2 B 4.0 m/s^2 C 6.0 m/s^2 D 8.0 m/s^2

6 On a distance–time graph, a horizontal line shows that the object is

A moving at constant speed B at rest
C accelerating D decelerating

7 What does the area under a speed–time graph represent?

A acceleration B speed
C distance travelled D time taken

8 Which statement about mass is correct?

A it is measured in newtons
B it is smaller on the Moon
C it is a measure of the quantity of matter
D it is a vector

9 An object of mass 5.0 kg is on a planet where g = 4.0 N/kg. What is its weight?

A 0.80 N B 1.25 N C 9.0 N D 20 N

10 A block measures 2.0 cm × 3.0 cm × 5.0 cm and has a mass of 240 g. What is its density?

A 0.125 g/cm^3 B 8.0 g/cm^3
C 48 g/cm^3 D 7200 g/cm^3

11 The water level in a measuring cylinder rises from 40 cm^3 to 55 cm^3 when a 45 g stone is added. What is the density of the stone?

A 0.82 g/cm^3 B 1.1 g/cm^3
C 3.0 g/cm^3 D 45 g/cm^3

12 A cyclist travels 600 m in 2.0 minutes. What is the average speed?

A 5.0 m/s B 300 m/s C 1200 m/s D 0.20 m/s

13 Which object floats in water (density 1.0 g/cm^3)?

A mass 50 g, volume 40 cm^3 B mass 30 g, volume 20 cm^3
C mass 40 g, volume 50 cm^3 D mass 90 g, volume 30 cm^3

14 What is the approximate acceleration of free fall near the Earth's surface?

A 1.6 m/s^2 B 9.8 m/s C 9.8 m/s^2 D 98 m/s^2

15 A ball is dropped from rest. Ignoring air resistance, what is its speed after 2.0 s?

A 4.9 m/s B 9.8 m/s C 20 m/s D 39 m/s

16 Which is equal to a density of 1 g/cm^3?

A 1 kg/m^3 B 10 kg/m^3 C 100 kg/m^3 D 1000 kg/m^3

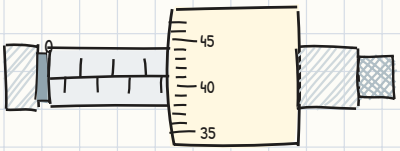
Warm-up: unit conversions *(answers at the back)*

1. 2.5 km = _____ m
2. 450 g = _____ kg
3. 3.5 min = _____ s
4. 72 km/h = _____ m/s
5. 250 cm^3 = _____ m^3
6. 3.0 mm = _____ m

Preview

PAPER 2 · STRUCTURED

1 A student uses a micrometer to measure the diameter of a wire, and times a pendulum.



(a) Write down the reading shown. [1]

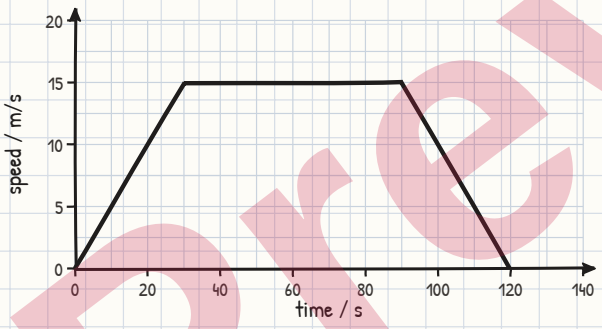
(b) With the jaws closed, the micrometer reads +0.03 mm. Calculate the corrected diameter. [1]

(c) Suggest why a micrometer is used rather than a 30 cm ruler. [1]

(d) The times for 20 oscillations are 31.2 s, 31.6 s and 31.4 s. Calculate the mean time and the period. [2]

(e) Explain why the student times 20 oscillations rather than one. [2]

2 The graph shows how the speed of a train changes between two stations.



(a) Describe the motion between 30 s and 90 s. [1]

(b) Calculate the acceleration in the first 30 s. [2]

(c) Calculate the distance between the stations. [3]

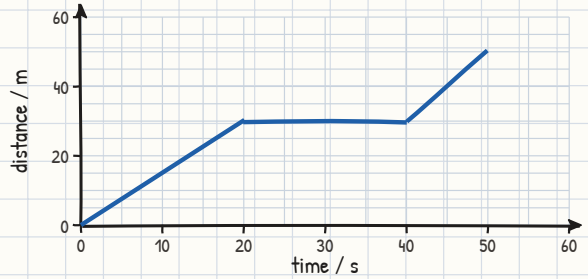
(d) Calculate the average speed for the journey. [2]

(e) On the graph, how would a larger deceleration be shown? [1]

Preview

PAPER 2 · STRUCTURED (CONTINUED)

3 The distance–time graph shows a student's walk.



(a) Calculate the speed during the first 20 s. [2]

(b) State what the student is doing between 20 s and 40 s. [1]

(c) Calculate the speed between 40 s and 50 s. [1]

(d) How does the graph show which part of the walk is fastest? [1]

4 An astronaut of mass 80 kg lands on Mars, where her weight is 296 N.

(a) Define *mass*. [1]

(b) Calculate the gravitational field strength on Mars. [2]

(c) State her mass on Mars. [1]

(d) Explain the difference between mass and weight. [2]

5 A student wants to find the density of a small, irregularly shaped stone.

(a) Describe how she should find the mass and the volume of the stone. [4]

(b) The stone has a mass of 65 g and a volume of 25 cm^3. Calculate its density in g/cm^3. [2]

(c) Convert your answer to kg/m^3. [1]

(d) Will the stone float in a liquid of density 2.9 g/cm^3? Explain. [1]

6 A boat points north and moves through the water at 4.0 m/s. The river flows east at 3.0 m/s.

(a) Calculate the size and direction of the resultant velocity of the boat. [3]

(b) State why velocity is a vector quantity. [1]

Quick-fire recall ✦ *P1 = 16 · P2 = 39*

1. Read a micrometer + correct a zero error.
2. 6 scalars and 8 vectors from the syllabus.
3. Define speed, velocity, acceleration.
4. 4 shapes each of d–t and v–t graphs.
5. Mass vs weight table; $W = mg$.
6. 3 methods for density.

Preview

Topic 1 · Part B

Forces & Momentum

pushes, pulls & Newton being right about everything

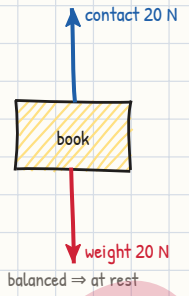
1.5 Forces — balanced & unbalanced

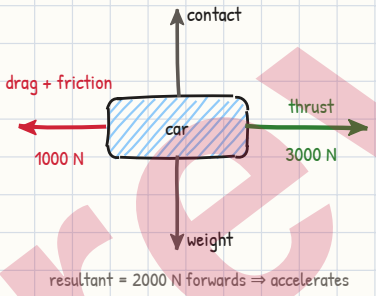
✱ Types of force [1.5.1.1]

force	what causes it / acts on
weight	gravitational pull of the Earth on a mass (downwards)
friction	surfaces rubbing; opposes sliding
drag / air resistance	moving through a liquid / air; opposes motion, grows with speed
tension	a stretched string, rope or spring (elastic force)
contact (normal) force	a surface pushing back, at 90° to it
thrust (driving force)	engine, rocket, push of a cyclist's legs
electrostatic	between charges
magnetic	between magnets / magnetic materials

✱ Free-body diagrams [1.5.1.2]

Draw the object as a box; draw **every force on it** as an arrow from the object, length showing size, labelled.





WE ✎ resultant along a line [1.5.1.5]

same direction: **add**. opposite directions: **subtract**; the resultant points the way of the bigger one.

car above: 3000 − 1000 = 2000 N forwards

tug-of-war: 450 N left, 520 N right ⇒ 70 N to the right

✱ What does the resultant force do?

resultant force	effect on the object
zero, object at rest	stays at rest
zero, object moving	constant speed in a straight line (constant velocity)
in the direction of motion	speeds up
opposite to the motion	slows down
at right angles to the motion	changes direction (e.g. circular motion)

Newton 1 in space Voyager 1 has coasted with its engines off for decades at about 17 km/s — no resultant force, so no change in velocity.

✱ Newton's first law [1.5.1.3]

"An object either remains at rest or continues to move in a straight line at constant speed unless acted on by a **resultant force**."

- forces **balanced** (resultant = 0) ⇒ no change in velocity: still stays still, moving keeps moving steadily
- forces **unbalanced** ⇒ velocity changes: it **speeds up, slows down or changes direction** [1.5.1.4]

✱ Resultant force = mass × acceleration [1.5.1.6]

$F = ma$ — F in N, m in kg, a in m/s²

1 newton is the force that gives a 1 kg mass an acceleration of 1 m/s². Bigger force ⇒ bigger acceleration; bigger mass ⇒ smaller acceleration.

WE 1 ✎ the car above

The car has mass 1250 kg. Find its acceleration.

$a = F \div m = 2000 \div 1250 =$ 1.6 m/s²

WE 2 ✎ a lift going up

A lift of mass 800 kg is pulled up by a cable with tension 9200 N. (g = 9.8 N/kg)

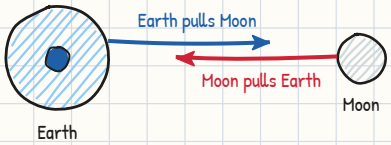
weight = 800 × 9.8 = 7840 N

resultant = 9200 − 7840 = 1360 N upwards

$a = 1360 \div 800 =$ 1.7 m/s² upwards

✱ Newton's third law [1.5.1.7–8]

"When object A exerts a force on object B, then object B exerts an **equal and opposite force** on object A."



The pair: **same size**, **opposite directions**, **same type** of force (here both gravitational), and act on **different objects** — so they never cancel.

classic trap ⚠

A book on a table: its **weight** and the **contact force** are equal and opposite — but they are **NOT** a third-law pair! They're different types and both act on the book. The pair of the book's weight is the book pulling the **Earth** up.

Preview

1.5.2 Friction, drag & terminal velocity

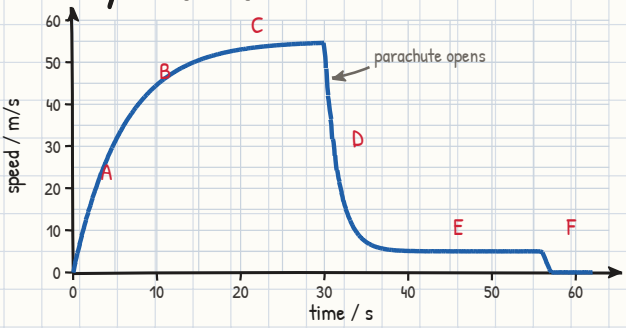
friction = a force that may **impede (oppose) motion** and **produce heating**. [1.5.2.1]

Rub your hands ⇒ warm. Brakes get **hot** (kinetic ⟶ thermal energy). Useful for grip and brakes; a nuisance for wear and wasted energy.

✱ A constant driving force, with and without drag [1.5.2.2]

- **no drag** (e.g. in a vacuum): resultant force stays the same ⇒ **constant acceleration** — speed keeps increasing
- **with drag**: drag **increases with speed** ⇒ resultant force falls ⇒ acceleration falls ⇒ eventually drag = driving force ⇒ **terminal velocity**

✱ The skydiver [1.5.2.3]



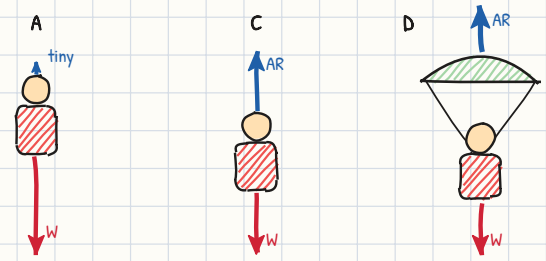
	forces	motion
A	weight > air resistance (tiny at first)	accelerates at about *g*
B	air resistance grows as speed grows	acceleration decreases (curve flattens)
C	air resistance = weight	**terminal velocity**, ≈ 55 m/s
D	parachute: air resistance > weight	decelerates
E	balanced again (at a lower speed)	new, lower terminal velocity ≈ 5 m/s

thinking distance ↑ if...	braking distance ↑ if...
higher **speed**	higher **speed** (rises very fast)
driver **tired**	heavier **load** (more mass)
alcohol or **drugs**	worn **tyres** (little tread)
distracted (phone!)	wet, icy or gravel **road**; worn brakes

driver factors ⇒ thinking distance; car & road factors ⇒ braking distance; speed ⇒ **both**.

model answer ✎ "explain terminal velocity" [4]

① at first weight > air resistance ⇒ resultant force down ⇒ **accelerates**
② as speed increases, **air resistance increases**
③ resultant force decreases ⇒ **acceleration decreases**
④ air resistance = weight ⇒ resultant = 0 ⇒ **terminal velocity**



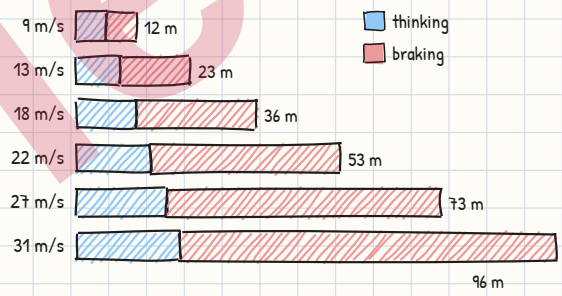
AR = air resistance (drag), W = weight

same idea, in a liquid 🧪

Drop a ball-bearing into a tall tube of oil: it speeds up briefly, then falls at a **constant terminal velocity** — time it between rubber bands spaced equally down the tube and the times become equal.

✱ Stopping distances [1.5.2.4–5]

thinking distance = distance travelled during the driver's **reaction time** (before the brakes go on)
braking distance = distance travelled **after the brakes are applied**, until it stops
stopping distance = thinking distance + braking distance



typical figures for a car on a dry road (UK Highway Code)

WE ✎ stopping in time?

A car travels at 20 m/s. Reaction time 0.70 s; braking distance 30 m.
thinking distance = speed × reaction time = 20 × 0.70 = **14 m**
stopping distance = 14 + 30 = **44 m**
after a drink, reaction time = 1.5 s ⇒ 20 × 1.5 = 30 m ⇒
stopping = **60 m**

speed kills ⚠

Thinking distance goes up in proportion to speed, but braking distance rises much faster — **double the speed ≈ four times the braking distance** (look at the red bars). That's why speed limits near schools are so low.

Preview

1.5.3 Elastic deformation – springs

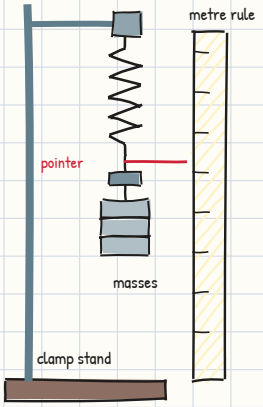
Forces can change the **size and shape** of an object – stretch, squash, bend or twist it. [1.5.3.1]

spring constant = force per unit extension [1.5.3.2]

$$\text{spring constant} = \frac{\text{force}}{\text{extension}} \qquad k = \frac{F}{x}$$ N/m or N/cm

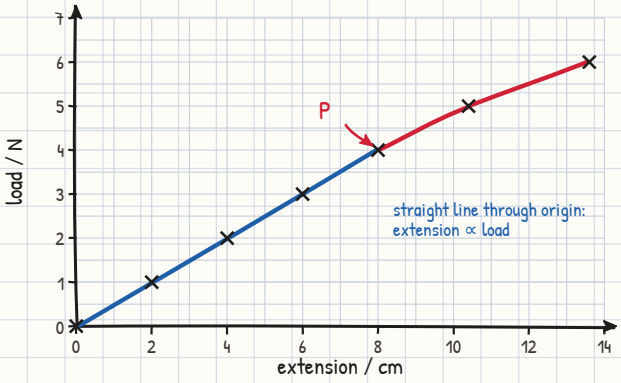
extension = stretched length – **original** length (not the total length!). A stiffer spring has a **bigger** k.

✱ The experiment [1.5.3.3]



1. measure the **original length** of the spring using the pointer and a vertical metre rule (eye level!)
2. add a 100 g mass (load ≈ 1 N); measure the new length
3. keep adding 100 g, recording length each time
4. extension = length – original length
5. plot **load / N** against **extension / cm**
6. safety: clamp the stand to the bench, keep feet away from falling masses, wear eye protection

load / N	0	1.0	2.0	3.0	4.0	5.0	6.0
length / cm	20.0	22.0	24.0	26.0	28.0	30.4	33.6
extension / cm	0	2.0	4.0	6.0	8.0	10.4	13.6



limit of proportionality (P) = the point beyond which the load–extension graph is **no longer a straight line** – extension is no longer proportional to load. [1.5.3.4]

WE 1 ✎ from the graph

gradient of straight part = load ÷ extension = 4.0 N ÷ 8.0 cm

k = 0.50 N/cm = 50 N/m (1 cm = 0.01 m)

WE 2 ✎ rearranging

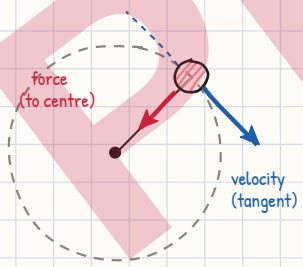
A spring has k = 25 N/m. A 2.0 N weight is hung on it. Find the extension.

$x = F \div k = 2.0 \div 25 =$ 0.080 m = 8.0 cm

graph axes trap ⚠

Here load is on the **y-axis**, so gradient = k. If a question plots **extension** on the y-axis, the gradient is **1/k** – check the axes!

1.5.4 Circular motion



Moving in a circle at **constant speed**: the **direction** keeps changing ⇒ velocity changes ⇒ there must be a **resultant force towards the centre**, **perpendicular to the motion**. [1.5.4.1] Cut the string and the ball flies off along the tangent (Newton 1).

if...	keeping constant...	then...
force ↑	mass, radius	**speed ↑**
force ↑	mass, speed	**radius ↓**
mass ↑	speed, radius	**force must ↑**

where the force comes from: car on a bend ⟶ **friction** · Moon & satellites ⟶ **gravity** · hammer-thrower ⟶ **tension** · electron round a nucleus ⟶ **electrostatic**. ($F = mv^2/r$ is not needed.)

WE ✎ the wet bend

Explain why a car going round a bend on an icy road may skid off the road.

① a force towards the centre is needed to keep the car moving in a circle – it comes from **friction** between tyres and road

② on ice friction is much smaller ⇒ not enough force for that speed and radius

③ it moves off along the tangent (Newton 1) – so **slow down**: less speed needs less force

"centrifugal force" doesn't exist 🎢

On a fast roundabout you feel "thrown outwards". There is no outward force! Your body's **inertia** tries to keep going straight, and the seat or wall has to push you **inwards** to make you go round.

spin dryer the drum pushes the clothes inwards, but water can pass through the holes – nothing pushes it inwards, so it carries on in a straight line and leaves the drum.

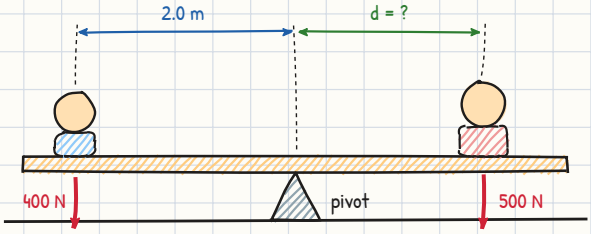
Preview

PREVIEW

1.5.5 Turning effect of forces — moments

The **moment** of a force measures its **turning effect** about a pivot. [1.5.5.1] Everyday examples: opening a door (handle far from the hinges), a long spanner on a tight nut, a seesaw, a wheelbarrow, scissors, a bottle opener.

moment = force × perpendicular distance from the pivot [1.5.5.2] unit: **N m**



principle of moments: for an object in **equilibrium**, the sum of clockwise moments = sum of anticlockwise moments about any point. [1.5.5.3]

Equilibrium also needs **no resultant force**: the pivot pushes up with 400 + 500 = 900 N.

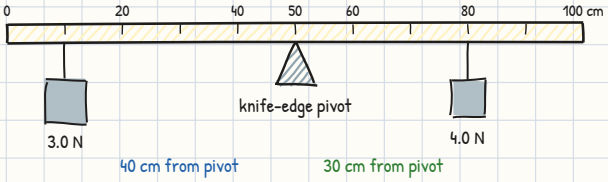
WE 1 ✎ balance the seesaw

anticlockwise moment = 400 × 2.0 = 800 N m

clockwise moment = 500 × d

balanced: 500d = 800 ⇒ d = 1.6 m — the heavier child sits **closer**

✱ Experiment to verify the principle [1.5.5.4]



1. balance a metre rule at its **centre** (50 cm) on a knife-edge pivot
2. hang masses on each side with loops of thread; slide them until the rule balances **horizontally**
3. record each force (weight) and its distance from the pivot
4. calculate: anticlockwise 3.0 × 0.40 = 1.2 N m; clockwise 4.0 × 0.30 = 1.2 N m ⇒ **equal** ✓
5. repeat with different masses & positions

WE 2 ✎ find the unknown weight

A rule pivoted at 50 cm balances with 2.0 N at the 10 cm mark and weight W at the 80 cm mark.

distances: 50 − 10 = 40 cm; 80 − 50 = 30 cm (measure from the **pivot**!)

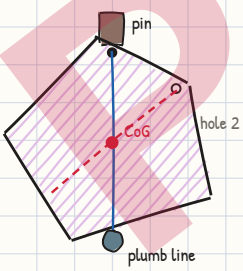
2.0 × 40 = W × 30 ⇒ W = 2.7 N

why door handles are where they are

Push near the hinge and you need a huge force; push at the handle (far away) and a small force gives the same moment. Same reason a long spanner undoes a stuck nut.

1.5.6 Centre of gravity

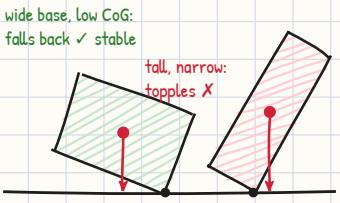
centre of gravity = the point through which the **whole weight** of an object appears to act. [1.5.6.1]



1. make 3 small holes near the edge of the card (lamina)
2. hang it from a pin through one hole so it **swings freely**; hang a **plumb line** from the pin
3. when still, mark the vertical line of the plumb line
4. repeat from another hole: the CoG is where the lines **cross** (the 3rd line checks it) [1.5.6.2]

CoG can be outside the object!

For a ring, a horseshoe or a boomerang, the centre of gravity is in the empty space in the middle. A high-jumper arching over the bar can get over while their CoG passes **under** it!



An object **topples** when the vertical line through its CoG falls **outside its base**. Low CoG + wide base = stable. [1.5.6.3]

examples: racing cars are low and wide · Bunsen burners and table lamps have heavy bases · double-decker buses are tilt-tested — passengers stand downstairs, not on top!

WE 3 ✎ weight of a uniform plank

A uniform plank 2.0 m long weighs 100 N and rests on a pivot 0.40 m from its left end. What downward force at the left end balances it?

uniform ⇒ weight acts at the **centre**, 1.0 m from the end = 0.60 m from the pivot

F × 0.40 = 100 × 0.60 ⇒ F = 150 N

Preview